Bishop’s University
Department of Computer Science
CS503 – Data Visualization
HW4 – Hyperspectral Data Visualization

Data: The Indian Pines hyperspectral (HS) image dataset consists of data collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. The spectral bands for this dataset are as follows:

- Visible (VIS): 0.4 - 0.7 micrometers (μm)
- Near-Infrared (NIR): 0.7 - 1.3 micrometers (μm)
- Short-Wave Infrared (SWIR): 1.3 - 2.5 micrometers (μm)

These bands cover a wide range of wavelengths and are used for various remote sensing applications, including agriculture, environmental monitoring, and land cover classification.

It covers an area in Indiana, USA, and consists of 145x145 pixels and 220 spectral bands. Each pixel in the image contains spectral information across these bands, providing detailed information about the surface materials present in the scene. The bands are stacked to form what we call HS cube shown in Figure 1.

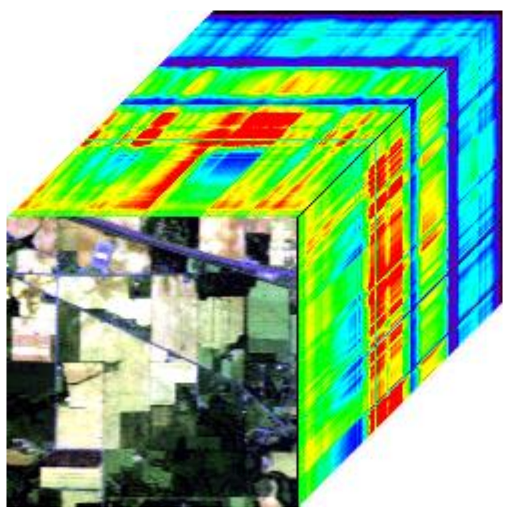
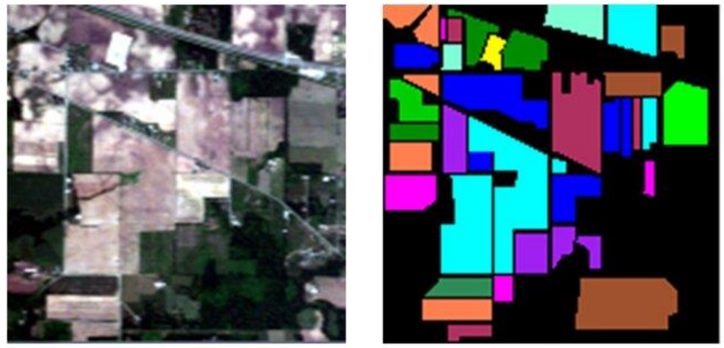


Figure 1. Indian Pines data cube (AVIRIS) [4].

Figure 2 displays the true color image of the Indian Pines scene alongside the corresponding reference image, often referred to as the ground truth image. The ground truth image delineates different classes, each with its assigned name and the number of samples, as detailed in Table 1.



(a) (b)

Figure 2. True color image of (a) Indian Pines scene and (b) the reference data (ground-truth) [1].

#	Class	Samples	Color
1	Alfalfa	46	
2	Corn-notill	1428	
3	Corn-mintill	830	
4	Corn	237	
5	Grass-pasture	483	
6	Grass-trees	730	
7	Grass-pasture-mowed	28	
8	Hay-windrowed	478	
9	Oats	20	
10	Soybean-notill	972	
11	Soybean-mintill	2455	
12	Soybean-clean	593	
13	Wheat	205	
14	Woods	1265	
15	Buildings-Grass-Trees-Drives	386	
16	Stone-Steel-Towers	93	
17	Uknown samples	10776	

Table 1. Ground-truth (reference) classes for the Indian Pines scene and their respective samples number [1]

The image data cube is stored in a ".mat" format. The ".mat" file format is commonly used to store data in MATLAB. It allows for the storage of variables, arrays, and other data structures in a structured format. In the context of the Indian Pines dataset, a ".mat" file may contain hyperspectral image data stored as MATLAB variables, along with any additional metadata or annotations. This format is widely used in scientific and engineering fields, especially for data interchange between MATLAB and other software environments like Python.

For more information about Indian Pines HS data, refer to [6].

Your task is:

1. The data is provided to you with some initial code in a Jupyter notebook, which you need to complete.
2. Load the HS data cube.
3. Pick three bands randomly and compose a false color image and display it.
4. Load the ground-truth image and display it.
5. Display the class distribution using a bar plot as follows:

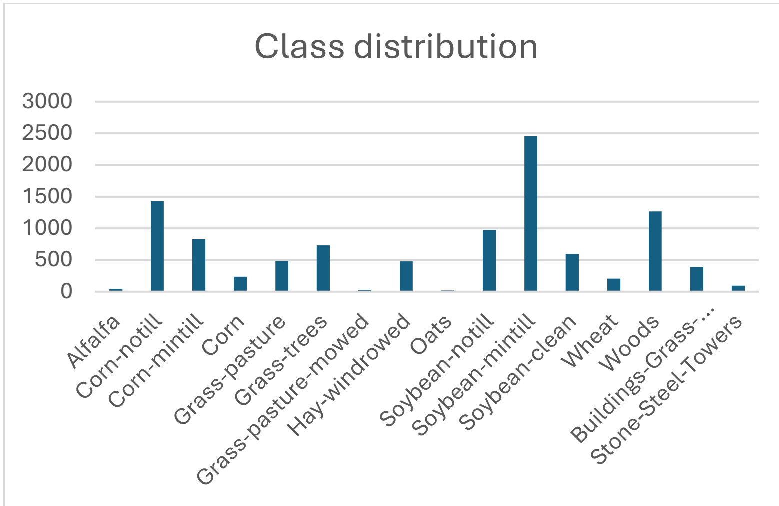


Figure 3. Class distribution generated by Office Excel.

6. Overlay the ground-truth as a mask on the HS image and display it alongside with the legend. Something like the following:

7.

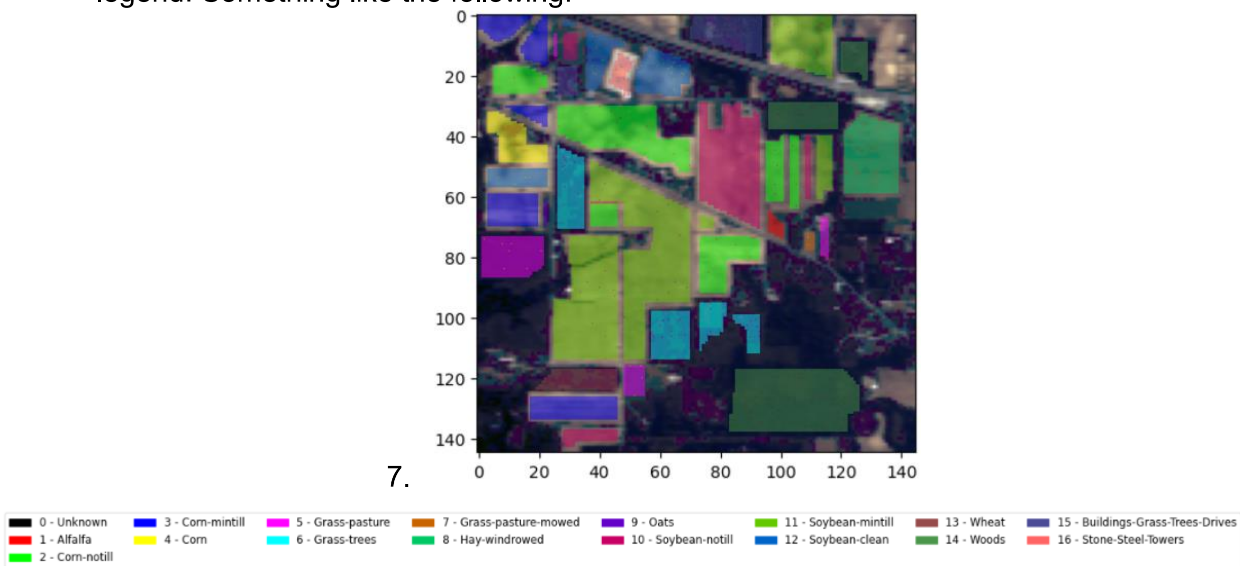


Figure 4. Overlaying ground-truth as mask on the HS image.

7. Select 1% of samples from each class and plot their mean vectors. These mean vectors represent the mean spectral signatures of the classes, as illustrated in Figure 5.
8. Apply PCA and t-SNE on the data cube and display the data scatter using the first two components of each dimensionality reduction (DR) method, as shown in Figure 6.

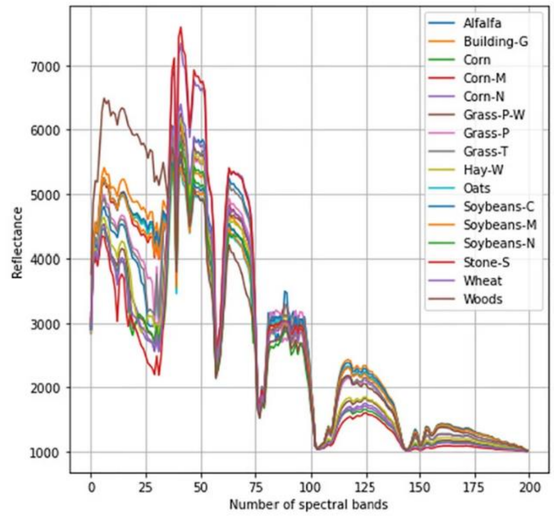
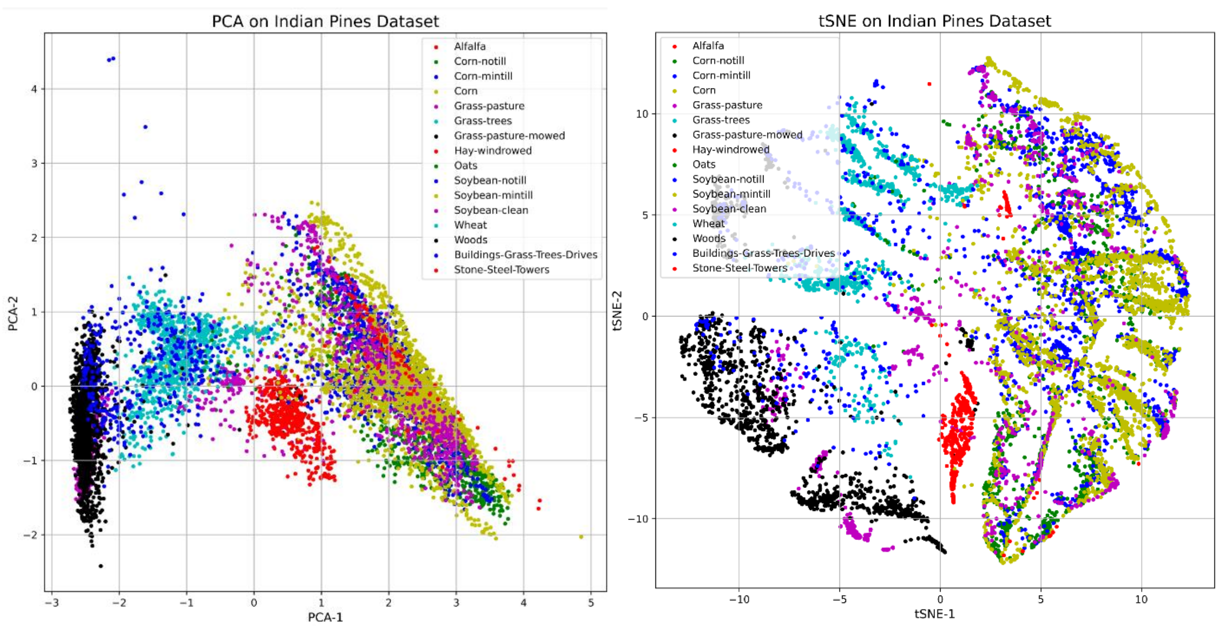


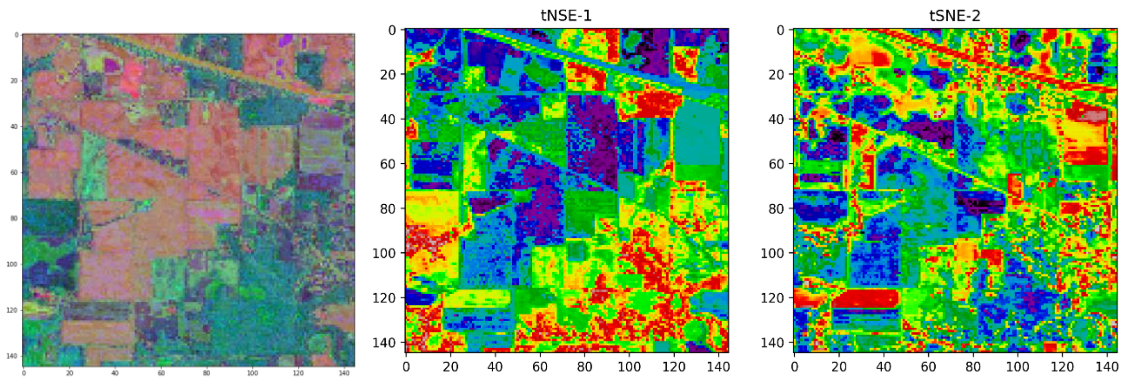
Figure 5. Mean spectral signatures of each class [2]



(a) (b)

Figure 6. Scatter plots of the DR algorithm features for Indian Pines dataset (**a**) PCA and (**b**) tSNE [3].

9. Reconstruct the image using the first important component of each DR method, as shown in Figure 7.



(a) (b) (c)

Figure 7. (a) Image reconstructed with PCA. (b) and (c) images reconstructed using two t-SNE's components [3].

Reference

- [1] S.Enayat Hosseini aria et al. http://dx.doi.org/10.1117/1.JRS.11.046010
- [2] Refka Hanachi et al. DOIhttps://doi.org/10.1007/s00521-023-09275-5
- [3] Md. Moazzem Hossain doi.org/10.3390/electronics12092082
- [4] Behnood Rasti DOI: 10.13140/RG.2.1.4097.4247
- [5] https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes